

Karan Choudhary

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EDUCATION

Michigan State University

Bachelor of Science, Computer Science, Minor in Cognitive Science, GPA: 3.97 / 4.0

East Lansing, MI

May 2026

EXPERIENCE

Undergraduate Learning Assistant - CSE440

January 2026 – May 2026

Michigan State University

East Lansing, MI

- Assisted students with various topics in artificial intelligence covering reinforcement learning to search algorithms
- Graded 200+ assignments, providing feedback in vital topics to help strengthen core fundamentals and overall understanding

IT Intern - Digital Innovation Lab

May 2025 – August 2025

Steelcase

Grand Rapids, MI

- Utilized Databricks and Tableau for data analysis, building data pipelines, and visualization, transforming complex datasets into clear, accessible insights for everyday users
- Collaborated cross-functionally with data science and data engineering teams to build production-ready tables and streamline code organization for deployment
- Leveraged Scikit-learn for machine learning technique, integrated OpenAI API and utilized PySpark and Pandas for efficient data aggregation and transformation

Software Engineering Fellow

August 2024 – September 2024

Headstarter

Remote

- Developed 5 projects using Next.js, OpenAI, Pinecone, and Stripe API, integrating various technologies
- Mentored by top industry professionals from Amazon, Bloomberg, and Capital One on Agile methodologies, CI/CD processes, Git version control, and microservice architectures

AI Intern

July 2024 – September 2024

Sports Media

Remote

- Utilized machine learning techniques in an audio processing project to enhance real-time voice communication through reduced latency and improved audio quality

PROJECTS

Generative Multi-Modal CAD AI Assistant | *Python, FastAPI, Next.js, LLMs, MCP, FreeCAD*

- Engineered a web application with 5 teammates to generate and modify CAD designs via LLMs, providing a multi-conversation interface and a design management system to revert to prior generated designs
- Integrated multimodal LLMs through a FastAPI backend and an MCP server bridging the LLM to FreeCAD, enabling model selection and design generation
- Implemented a multimodal input pipeline and design downloading system, enabling users to download CAD designs at any point in the conversation

Movie Recommender Website | *Python, Streamlit, Scikit-learn, NumPy, Pickle*

- Constructed a movie recommender system wielding Streamlit, enhancing deployment efficiency and accomplishing a 93% user satisfaction rate in recommendation relevance
- Administered Scikit-learn's cosine similarity and TfidfVectorizer for feature vectorization in movie recommendation system reducing computational runtime by 63%

Parking Lot Predictor | *Python, OpenCV, Pickle, NumPy*

- Architected an automated parking lot monitoring system incorporating Python and OpenCV, discerning between occupied and vacant parking spaces through analysis of video feeds resulting in greater accuracy by 35%

TECHNICAL SKILLS & INVOLVEMENT

Languages: Python, SQL, C++, HTML, CSS, JavaScript

Developer Tools: Git, GitLab, GitHub, AWS, Linux, Databricks, Tableau, Azure DevOps, Postman, FreeCAD

Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib, OpenCV, BeautifulSoup, Pickle, Requests, Keras, PySpark, Tqdm, Pydantic, JSON, OpenAI, FastAPI, Next.js

Organizations: EV Scholars, MSU AI Club